

Consider a **free massive particle**: a particle with no external interactions. Let m denote its Newtonian mass, t the time measured in an inertial laboratory frame, and $\mathbf{v} = d\mathbf{r}/dt$ its velocity in that frame. We will use ordinary Lagrangian mechanics throughout.

1. The action and the Lagrangian

Start with the assumed free-particle action

$$S = \int_{\tau_1}^{\tau_2} \alpha d\tau,$$

where τ is **proper time**, the time measured by a clock moving with the particle, and α is a constant characterizing the particle.

This form expresses the absence of external interactions: there are no external fields or position-dependent terms in the action. Proper time is independent of the inertial frame used to describe the motion, so this action also has the same value in every inertial frame. Its normalization, including the sign of α , remains to be determined.

The Lorentz transformation gives, along the particle's motion,

$$dt = \gamma d\tau, \quad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}},$$

where $v = |\mathbf{v}|$. Thus

$$S = \int_{t_1}^{t_2} \alpha \sqrt{1 - \frac{v^2}{c^2}} dt.$$

By the definition $S = \int L dt$, the Lagrangian is

$$L = \alpha \sqrt{1 - \frac{v^2}{c^2}}.$$

2. Derive the momentum from the exact Lagrangian

In Lagrangian mechanics, the momentum conjugate to x_i is defined by

$$p_i = \frac{\partial L}{\partial v_i}.$$

To see why this is the relevant conserved quantity, use the Euler–Lagrange equation:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial v_i} \right) = \frac{\partial L}{\partial x_i}.$$

Our free-particle Lagrangian has no position dependence, so

$$\frac{dp_i}{dt} = 0.$$

Thus $\partial L / \partial v_i$ is the conserved momentum associated with motion in the i -direction.

Since $v^2 = v_x^2 + v_y^2 + v_z^2$, differentiate the exact Lagrangian with respect to each velocity component:

$$\begin{aligned} p_i &= \frac{\partial L}{\partial v_i} \\ &= \frac{\alpha}{2} \left(1 - \frac{v^2}{c^2} \right)^{-1/2} \left(-\frac{2v_i}{c^2} \right) \\ &= -\frac{\alpha v_i}{c^2 \sqrt{1 - v^2/c^2}}. \end{aligned}$$

Therefore,

$$\mathbf{p} = -\frac{\alpha}{c^2} \gamma \mathbf{v}, \quad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

3. Determine α by matching the low-speed momentum

For $v \ll c$, expand the exact momentum:

$$\mathbf{p} = -\frac{\alpha}{c^2} \left[1 + \frac{v^2}{2c^2} + O\left(\frac{v^4}{c^4}\right) \right] \mathbf{v} \approx -\frac{\alpha}{c^2} \mathbf{v}.$$

In the Newtonian limit, this must reproduce the classical momentum

$$\mathbf{p}_N = m \mathbf{v}.$$

Comparing coefficients,

$$-\frac{\alpha}{c^2} = m, \quad \alpha = -mc^2.$$

The matching requires agreement with the **normalization** of classical momentum, as well as its conservation: conservation alone cannot determine α , because multiplying a conserved quantity by any constant preserves its conservation.

Substituting $\alpha = -mc^2$ into the exact Lagrangian and momentum,

$$\begin{aligned} L &= -mc^2 \sqrt{1 - \frac{v^2}{c^2}}. \\ \mathbf{p} &= \gamma m \mathbf{v}. \end{aligned}$$

The low-speed approximation has served only to determine α . We now use the exact expression.

For one-dimensional motion, this reads $p = \gamma mv$. As $v/c \rightarrow 0$, $\gamma \rightarrow 1$, and the result reduces to $p = mv$.

4. Derive the energy H

In Lagrangian mechanics, define

$$H = \sum_i p_i v_i - L = \mathbf{p} \cdot \mathbf{v} - L.$$

Its conservation follows directly from the Euler–Lagrange equations. Along the motion,

$$\begin{aligned} \frac{dH}{dt} &= \sum_i (\dot{p}_i v_i + p_i \dot{v}_i) - \sum_i \left(\frac{\partial L}{\partial x_i} v_i + \frac{\partial L}{\partial v_i} \dot{v}_i \right) - \frac{\partial L}{\partial t} \\ &= -\frac{\partial L}{\partial t}. \end{aligned}$$

Our Lagrangian has no explicit time dependence, so $dH/dt = 0$. This is the conserved energy.

Using the momentum and Lagrangian already obtained,

$$\begin{aligned} H &= \gamma m v^2 + m c^2 \sqrt{1 - \frac{v^2}{c^2}} \\ &= \gamma m v^2 + \gamma m c^2 \left(1 - \frac{v^2}{c^2} \right) \\ &= \gamma m c^2. \end{aligned}$$

Hence

$$H = \gamma m c^2.$$

At rest, this gives

$$H(0) = m c^2.$$

The kinetic energy is the increase above the rest energy:

$$T = H - H(0) = (\gamma - 1) m c^2.$$

For $v \ll c$,

$$T \approx \frac{1}{2} m v^2,$$

which recovers the Newtonian result.